Current density

Dr. Sampath Kumar

Associate Professor

Department of ECE

MIT, Manipal

Problem:

Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z > 0$, $\varepsilon_{r1} = 4$ and for $z < 0$, $\varepsilon_{r2} = 3$. A uniform electric field $\mathbf{E}_1 = 5\mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z$ kV/m exists for $z \geq 0$. Find

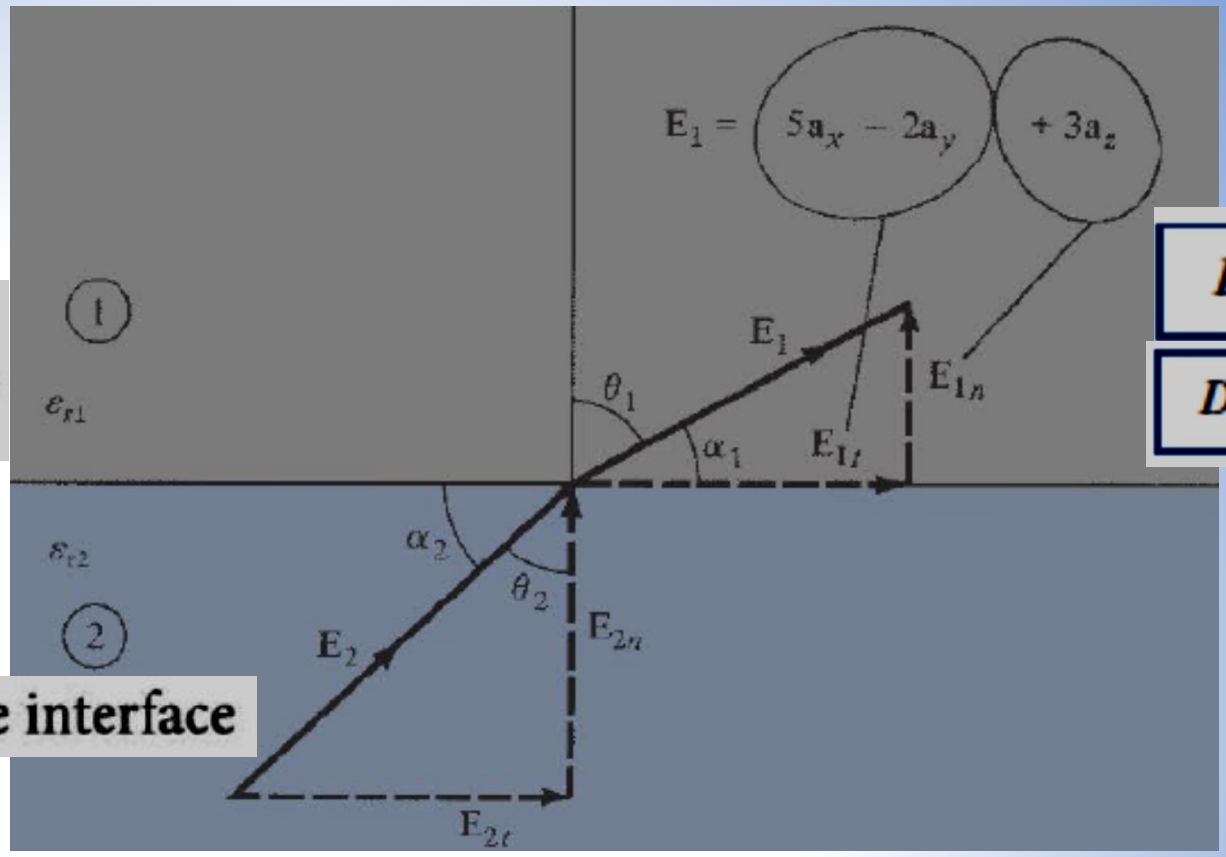
(a) $\mathbf{E}_2$ for $z \leq 0$
(b) The angles $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the interface
(c) The energy densities (in J/m^3) in both dielectrics
(d) The energy within a cube of side 2 m centered at $(3, 4, -5)$

(a) $\mathbf{E}_2$ for $z \leq 0$

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} = 5\mathbf{a}_x - 2\mathbf{a}_y$$

$$\mathbf{E}_{2n} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}\mathbf{E}_{1n} = \frac{4}{3}(3\mathbf{a}_z) = 4\mathbf{a}_z$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}$$
$$= 5\mathbf{a}_x - 2\mathbf{a}_y + 4\mathbf{a}_z \text{ kV/m}$$



$$E_{\tan 1} = E_{\tan 2}$$

$$D_{N1} - D_{N2} = \rho_S$$

(b) The angles $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the interface

$$\alpha_1 = 90 - \theta_1$$
$$\alpha_2 = 90 - \theta_2$$

$$\tan\theta_1 = \frac{E_{1t}}{E_{1n}} = \frac{\sqrt{29}}{3} = 1.795 \rightarrow \theta_1 = 60.9°$$

$$\alpha_1 = 29.1°$$

$$\tan\theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{\sqrt{29}}{4} = 1.346 \rightarrow \theta_2 = 53.4°$$

$$\alpha_2 = 36.6°$$

Verify $\dfrac{\tan\theta_1}{\tan\theta_2} = \dfrac{\epsilon_1}{\epsilon_2}$

(c) The energy densities are given by $\mathbf{E}_1 = 5\mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z \text{ kV/m}$

$$w_{E1} = \frac{1}{2}\varepsilon_1|\mathbf{E}_1|^2 = \frac{1}{2} \times 4 \times \frac{10^{-9}}{36\pi} \times (25 + 4 + 9) \times 10^6$$
$$= 672\ \mu\text{J/m}^3$$

$$w_{E2} = \frac{1}{2}\varepsilon_2|\mathbf{E}_2|^2 = \frac{1}{2} \times 3 \times \frac{10^{-9}}{36\pi} (25 + 4 + 16) \times 10^6$$
$$= 597\ \mu\text{J/m}^3$$

(d) The energy within a cube of side 2 m centered at $(3, 4, -5)$

(d) At the center $(3, 4, -5)$ of the cube of side 2 m, $z = -5 < 0$; that is, the cube is in region 2 with $2 \le x \le 4, 3 \le y \le 5, -6 \le z \le -4$. Hence

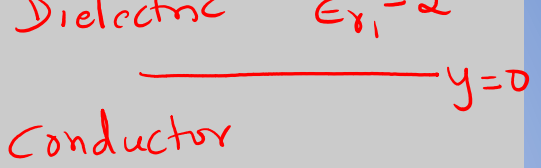
$$W_E = \int w_{E2}\, dv = \int_{x=2}^{4} \int_{y=3}^{5} \int_{z=-6}^{-4} w_{E2}\, dz\, dy\, dz$$

$$= w_{E2}(2)(2)(2)$$
$$= 597 \times 8\ \mu\text{J} = 4.776\ \text{mJ}$$

Pb: ②

Region $y < 0$ consists of a perfect conductor while region $y > 0$ is a dielectric medium ($\varepsilon_{1r} = 2$) as in ~~Figure 5.16~~. If there is a surface charge of 2 nC/m^2 on the conductor, determine **E** and **D** at

(a) $A(3, -2, 2)$
(b) $B(-4, 1, 5)$



(a) Point $A(3, -2, 2)$ is in the conductor since $y = -2 < 0$ at A. Hence,

$$\mathbf{E} = \mathbf{0} = \mathbf{D}$$

(b) Point $B(-4, 1, 5)$ is in the dielectric medium since $y = 1 > 0$ at B.

$$D_n = \rho_S = 2\,\text{nC/m}^2$$

$$\mathbf{D} = 2\mathbf{a}_y\,\text{nC/m}^2$$

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon_o \varepsilon_r} = 2 \times 10^{-9} \times \frac{36\pi}{2} \times 10^9 \mathbf{a}_y = 36\pi \mathbf{a}_y$$

$$= 113.1\,\mathbf{a}_y\,\text{V/m}$$

D5.9. Let Region 1 ($z < 0$) be composed of a uniform dielectric material for which $\epsilon_r = 3.2$, while Region 2 ($z > 0$) is characterized by $\epsilon_r = 2$. Let $\mathbf{D}_1 = -30\mathbf{a}_x + 50\mathbf{a}_y + 70\mathbf{a}_z$ nC/m^2 and find: (*a*) D_{N1}; (*b*) $\mathbf{D}_{t1}$; (*c*) D_{t1}; (*d*) D_1; (*e*) θ_1; (*f*) $\mathbf{P}_1$.

Ans. (*a*) 70 nC/m^2; (*b*) $-30\mathbf{a}_x + 50\mathbf{a}_y$ nC/m^2; (*c*) 58.3 nC/m^2; (*d*) 91.1 nC/m^2; (*e*) 39.8°; (*f*) $-20.6\mathbf{a}_x + 34.4\mathbf{a}_y + 48.1\mathbf{a}_z$ nC/m^2

D5.10. Continue Problem D5.9 by finding: (*a*) $\mathbf{D}_{N2}$; (*b*) $\mathbf{D}_{t2}$; (*c*) $\mathbf{D}_2$; (*d*) $\mathbf{P}_2$; (*e*) θ_2.

Ans. (*a*) $70\mathbf{a}_z$ nC/m^2; (*b*) $-18.75\mathbf{a}_x + 31.25\mathbf{a}_y$ nC/m^2; (*c*) $-18.75\mathbf{a}_x + 31.25\mathbf{a}_y + 70\mathbf{a}_z$ nC/m^2; (*d*) $-9.38\mathbf{a}_x + 15.63\mathbf{a}_y + 35\mathbf{a}_z$ nC/m^2; (*e*) 27.5°

P = polarization

Current and Current Density

- Electric charges in motion constitute a *current*.
- The unit of current is the ampere (A), defined as a rate of movement of charge passing a given reference point (or crossing a given reference plane).

$$I = \frac{dQ}{dt}$$

- Current is *defined* as the motion of positive charges, although conduction in metals takes place through the motion of electrons.
- A new quantity, current density J is defined. It is measured in amperes per square meter (A/m^2).
- Through integration, the total current is obtained:

$$I = \int_S \mathbf{J} \cdot d\mathbf{S}$$

D5.1. Given the vector current density $\mathbf{J} = 10\rho^2 z\mathbf{a}_\rho - 4\rho\cos^2\phi\,\mathbf{a}_\phi$ mA/m^2: (*a*) find the current density at $P(\rho = 3, \phi = 30°, z = 2)$; (*b*) determine the total current flowing outward through the circular band $\rho = 3, 0 < \phi < 2\pi, 2 < z < 2.8$.

Ans. (*a*) $180\mathbf{a}_\rho - 9\mathbf{a}_\phi$ mA/m^2; (*b*) 3.26 A

Hint: ρdϕdz **aρ**

Current and Current Density

$$\mathbf{J} = \rho_v \mathbf{v}$$

■ J = ρ_vv is then called *convection current density*.

V = velocity of charge density

Also, $\mathbf{J} = \sigma \mathbf{E}$ σ = conductivity

$$I = \oint_S J \cdot dS = -\frac{dQ_i}{dt}$$

• **The Integral Form of the Continuity Equation**

$$\int_{vol} (\nabla \cdot \mathbf{J}) dv = -\frac{d}{dt} \int_{vol} \rho_v dv$$

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

• **The Differential Form (Point Form) of the Continuity Equation**


Thank
you